

Tissue

Tissues are groups of cells that have a similar structure and act together to perform a specific function.

There are four types of tissues as:

- i- Epithelial Tissue
- ii- Connective Tissue
- iii- Nervous Tissue
- iv- Muscular Tissue

Epithelial Tissue:

A group of cells covering the skin, cavities, tubes and organs of the body is known as **Epithelium Tissue**.

Epithelial tissues are widespread throughout the body. They form the covering of all body surfaces, line body cavities and hollow organs, and are the major tissue in glands.

Functions:

They perform a variety of functions that include protection, secretion, absorption, excretion, filtration, diffusion, and sensory reception. Epithelial layers contain no blood vessels, so they must receive nourishment via diffusion of substances from the underlying connective tissue.

There are two types of epithelial tissue as:

- i- Simple epithelial
- ii- Stratified epithelial

Simple epithelial:

Simple epithelial consists of single layer of cells. It is usually found on absorptive or secretory surfaces.

There are four types of simple epithelial tissue as:

- i- Simple Squamous Epithelial
- ii- Simple Cuboidal Epithelial
- iii- Simple Columnar Epithelial

iv- Pseudo Stratified Columnar Epithelial

Simple Squamous Epithelial:

Simple squamous epithelial is a single layer of flat cells in contact with the basal lamina (one of the two layers of basement membrane) of the epithelium.

This type of **epithelium** is often permeable and occurs where small molecules need to pass quickly through membranes via filtration or diffusion.

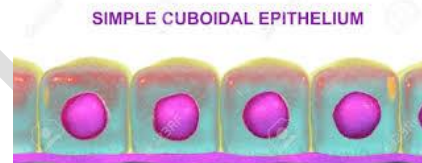
Their width is greater than height.



Simple Cuboidal Epithelial:

These are cube shaped and single layer

Layer of cuboid cell. These are found in



Organs that are specialized for secretion, such as salivary glands and thyroid follicles, and those are specialized for diffusion, such as kidney tubules. Their height is equals to width.

Simple Columnar Epithelial:

A simple columnar epithelial is a columnar

Epithelium that is uni-layered. In humans,

a simple columnar epithelium lines most



organs of digestive tract including stomach, small intestine, and large intestine.

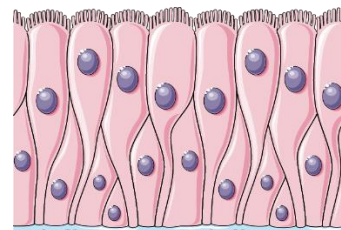
Simple columnar epithelium line Uterus. Their height is greater than width.

Pseudo Stratified Columnar epithelial:

These consists of single cell layer but looks likes

multi-cell layered. These are found in lining of

the trachea as well as upper respiratory tract.



Stratified Epithelial

Stratified epithelial consists of several layers of cells of various shapes. The main function of stratified epithelial is to protect underlying structure.

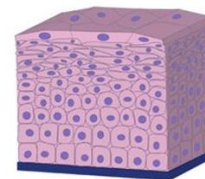
There are four types of stratified epithelial as:

- i- Stratified squamous epithelial
- ii- Stratified cuboidal epithelial
- iii- Stratified columnar epithelial
- iv- Transition epithelial

Stratified Squamous Epithelium:

These are multi-layered of flat cells.

Their width is greater than height.



Stratified Squamous

Stratified Cuboidal Epithelial:

It is a type of epithelial tissue composed of multi-layered of cube-shaped cells.

Their height is equals to width.



Stratified Columnar Epithelial:

It is a epithelium that composed of column shaped cells arranged in multi-layered. These are found in conjunctiva of eye, in parts of pharynx, anus, uterus and male urethra.

Transitional Epithelium:

- ⇒ Changeable
- ⇒ Protection
- ⇒ That changes upon tension

Squamous => Cuboid => Columnar.

Connective Tissue

Tissue that connects, binds, supports and protects other tissues or organs is called connective tissue.

Functions:

- 1- Bind and support
- 2- Protection
- 3- Insulation
- 4- Storing reserve fuel
- 5- Transportation

There are two types of connective tissue as:

- i- Loose connective tissue
- ii- Dense connective tissue

Loose Connective Tissue:

The tissue in which fibers are less than ground substances called loose connective tissue. For example: Areolar Tissue.

Dense Connective Tissue:

The tissues in which fibers are greater than ground substances called dense connective tissue. For example: Tendons and Ligaments.

Tendons: are muscle to bone.

Ligaments: are bones to bones.

Components of connective tissue:

- i- Cells
- ii- Fibers (Thread like structure)
- iii- Ground Substances

Matrix

Types Connective Tissue Cells:

- 1- Fibroblast
- 2- Adipocyte
- 3- Macrophages



- 4- Mast cells
- 5- leucocytes

Fibroblast:

- ⇒ Extracellular
- ⇒ Production of matrix

Adipocyte:

- ⇒ Fat cells
- ⇒ Adipose tissue

Macrophages:

- ⇒ Is the type of immune system
- ⇒ Engulfs and digest extracellular debris
- ⇒ Eat and digest particles

Mast Cells:

- ⇒ Protection
- ⇒ Wound Healing
- ⇒ Rich in Histamines

Leucocytes:

- ⇒ WBC's Protection
- ⇒ Deals with infectious diseases

Fibers

The intercellular matrix of **loose connective tissue** is composed of various fibers and **ground substance**. There are three types of fibers as:

Elastic Fibers:

- Soft
- Lung, artery

Collagen Fibers:

- Hard
- Ligament, Tendons

Reticular Fibers:

- Soft
- Skeleton, liver, bone marrow, lymph-nodes

Ground Substance

It is a gel like material in the extracellular space tall the components of extracellular molecules.

OR

It is an amorphous gel like material in the extracellular matrix except fibers is known as ground substance.

⇒ **Support + lubricate + development + collagen + tissue**

Nervous Tissue (Neuron)

Nerve cells, is also called neurons are the basic structural and functional unit of nervous tissue. They are responsible for conducting nerve impulses from one part of the body to another. They are the basic information processing unit of nervous system. The human nervous system contains about 12 Trillion neuron.

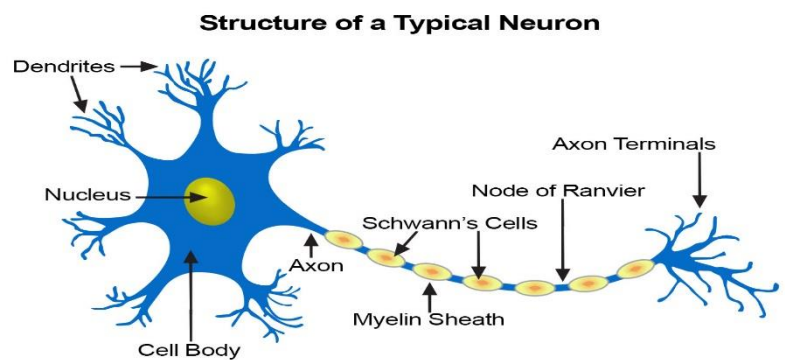
OR

Neuron is the functional unit of nervous tissue that receives processes and transmits information through electrical and chemical signals.

Structure of Neuron:

There are following parts in neuron as:

- 1- Dendrites
- 2- Cell body
- 3- Nucleus
- 4- Axon
- 5- Myelin Sheath
- 6- Schwann cells
- 7- Nodes of Ranvier
- 8- Synapse



Cell Body (soma or perikaryon):

- 1- Processes
- 2- Controls metabolism of the cell

The nucleus-containing central part of a neuron exclusive of its axons and dendrites — also called perikaryon, soma.

Dendrites:

It collects information from other cells and transmit to cell body.

Nodes of Ranvier:

It allows rapid conduction of impulses / signals forcing them to jump from one node to the next node.

Axon (Nerve Fiber):

This is the name given to the longest process of the neuron and is also called nerve fiber. Axon carry impulses away from the nerve cell body to the other neuron, or muscular or glandular tissue. The axon emerges from the cell body at the region termed the **AXON HILLOCK**.

Myelin sheath (layer of fatty tissue):

The insulation around the axon to prevent the disturbance / distortion in the conduction of signals.

Size:

Axon are variable in length from a few millimeters (in brain) to a meter or more between the spinal cord and toes.

a. Axon collaterals:

Axon do not branch as much as dendrites but many axons along their course show side branchings called collaterals.

b. Axon terminalis:

The axons and axon collaterals terminate by branching into many fine filaments known as axon terminalis.

Synapse:

A synapse is a structure that permits a neuron to pass an electrical or chemical signals to the other neuron.

OR

A specialized structure between the neuron is called synapse.

Types:

There are three types of neuron as:

- i- Sensory neuron or Afferent neuron
- ii- Motor neuron or Efferent neuron
- iii- Intermediate neuron or Relay neuron

Sensory neuron:

These type of neurons receive afferent fibers, responsible for various sensations. For Example: pain, touch, pressure, light, sound, smell, and taste neurons.

Motor neuron:

These neuron send impulses from the brain and spinal cord to the effectors, which may be muscles or glands.

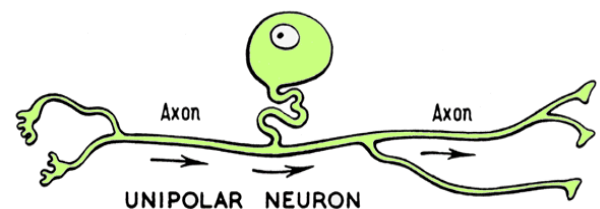
Classification of Neurons:

The different neuron in the body are classified on the basis of their structure and function.

Histological classification of neurons

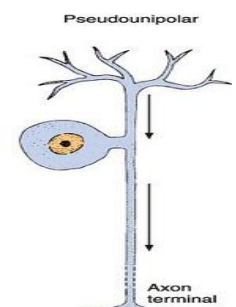
a. Uni-polar neurons (mono-polar):

This type neuron have only one process
Which functionally an axon. In man these
Are only in one place as mesencephalic
Nucleus of the terminal nerve.



b. Pseudo unipolar neurons:

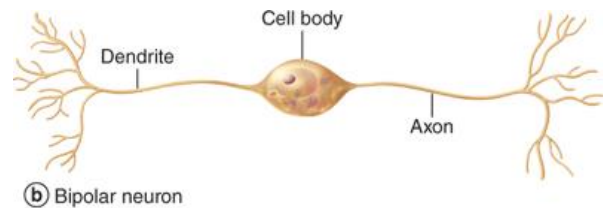
In this type of neuron, the axon and dendrites arise from a



common stem that divides into two processes, one acting as a dendrite and the other as the axon. Example is posterior spinal root ganglia.

c. Bipolar neurons:

In this type of neuron there is one Axon and one dendrite, each arising At different sites of the cell body



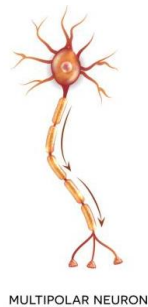
Opposite to each other and having the same size. Example: Retina of eye, olfactory mucous membrane, and inner ear.

d. Multipolar neurons:

These neurons have large number of dendrites arising from cell body. The dendrite may arise from one pole of the cell body or may arise from all parts of cell body. There is only one long process that represents the axon, most neuron of the CNS are of this type. Example: Brain and Spinal Cord.

e. Neuron without axon:

A few neuron in the CNS processes dendrites but no axon. In these cells nervous connections occur in any direction along their dendrite-like processes. Example: Amacrine cells in retina.



Muscular Tissue

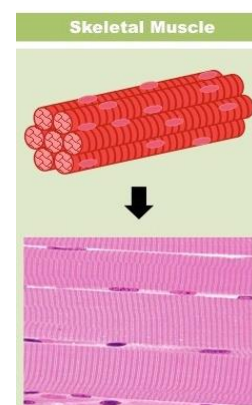
It is a soft tissue that composes muscle in body and gives the ability for muscle contraction.

There are three types of muscular tissue as:

- 1- Skeletal Muscle Tissue
- 2- Smooth Muscle Tissue
- 3- Cardiac Muscle Tissue

Sarcomere:

It is the basic functional unit of muscle tissue.

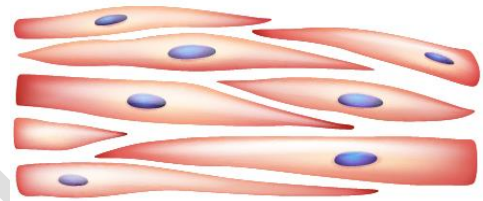


Skeletal Tissue:

- ⇒ These are striated in structure.
- ⇒ These are under voluntary control.
- ⇒ The maximum length is 35 cm.
- ⇒ Tendons, ligament
- ⇒ Cylindrical shaped
- ⇒ **Function:** locomotion, movement, posture

Smooth Tissue:

- ⇒ These are non-striated in structure.
- ⇒ These are involuntary (not in our control).
- ⇒ Short, soft
- ⇒ In esophagus, bladder, bronchi, vessel etc.
- ⇒ **Functions:** internal organs



Smooth muscle

Cardiac Tissue:

- ⇒ These are semi-striated in structure.
- ⇒ Involuntary
- ⇒ These are in branch form.
- ⇒ In heart (heart beat)

Functions:

- Self-contracting
- Rhythmic manner

